Errors Are Not Choke Points: Counterfactual Localisation of LLM Reasoning Failures, and Why Restart Is Hard to Beat

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Abstract

Process reward models are trained and benchmarked to find the first incorrect step of a reasoning trace, and verifier-guided search returns to that step to repair it. This assumes that a failure is *decided* at the labelled error, and that resampling from just before it beats starting again. We test both assumptions with one measurement, v_k: the generator's own probability of success from prefix k, estimated from 125,832 on-policy continuations of human-annotated failed traces from four generators.

Both assumptions mostly fail. 33–66% of failures are decided before the first step, and in the rest the human-labelled error is the decisive step only 0.438–0.647 of the time, against 0.16–0.20 by chance (preregistered; confirmed on a held-out model family). On ChokeBench, a benchmark and evaluation protocol we release (230 failed traces, four generators), a process reward model finds 73% of human first errors but only 45% of the decisive steps, and on held-out models ChokeBench scores rank localisers by their actual recovery gain better than first-error scores do (Spearman +0.52 against +0.22). At matched compute, no deployable resampling point reliably beats plain restart, and even a free oracle for the decisive step gains only +0.028 at four restarts' budget, while restarting recovers 44–66% of failed solutions within four attempts. A supermartingale bound explains the pattern: a failed on-policy prefix is, in expectation, worth no more than a fresh start, so localising the failure saves tokens rather than adding successes. On exactly on-policy traces of a reasoning model (DeepSeek-R1-Distill-1.5B), no cut of a failed trace is worth more than a fresh start, as the bound predicts, though a gain from cheaper late attempts could not be ruled out; most of its failures are budget exhaustion, with no wrong step to find. As a negative result, early-warning signals mostly read problem difficulty: the question and model alone give AUROC 0.588, and the first quarter of the reasoning adds +0.005 (n.s.).

In a secondary study on retrieval-augmented generation (RAGTruth, Llama-2-7B), a preregistered detect–resample controller lowers the judged hallucination rate against the base model (-0.047, $p = 0.0003$), and plain restarting does about the same (-0.042 [-0.063, -0.022], exploratory). That judge agrees only weakly with people ($\kappa = 0.219$), the choke-point-aware variant is no better than plain restarting (-0.005) at higher cost (1.77× against 1.36× the base model), and outputs have not yet been rated by people. The measured gain also depends on the grader: +0.326 under a grader trained like the gate's detector, +0.017 (n.s.) under an independent judge. Counterfactual resampling itself is not new; what we add is its comparison with human error labels, verifiers and recovery policies at matched compute. All rollouts, preregistrations and the code behind every number are released.

1 Introduction

Language models fail in the middle of a generation: a reasoning step goes wrong, or an answer drifts away from its source. Most mitigation methods are *surgical*. Process reward models (PRMs) are trained and benchmarked to find the *first incorrect step* of a reasoning trace [2, 3, 5], and verifier-guided search backtracks to it. Early-warning detectors predict from a partial trace that the answer will be wrong [9, 10]. Retrieval-augmented systems detect a hallucinated sentence and rewrite or regenerate from there [33, 32]. Surgery is attractive because it promises to keep the good part of a generation and pay only for the bad part. It rests on one assumption: a failure is *decided* at an identifiable step, and resampling from just before that step beats resampling from scratch.

This paper tests that assumption directly, with one quantity throughout. For a trace written by model M, let v_k be M's probability of reaching a correct (or faithful) output from the prefix ending at step k, estimated by sampling M's own continuations. v_0 is the from-scratch solve rate, so *restart* is resampling from $k = 0$ and *backtracking* to step j is resampling from $k = j - 1$. A *choke point* is where v collapses; an *error* is where a human annotator says the reasoning first goes wrong.

The assumption mostly fails. Many failures are decided before the first step, verifiers find the human label rather than the choke point, and even a free oracle for the choke point gains little over restart at matched compute, because a failed on-policy prefix is on average worth no more than a fresh start (Proposition 1 (Appendix A)). Restarting itself recovers 44–66% of failed maths solutions within four attempts' compute. A secondary study on RAG asks whether the same holds for hallucinations: there, plain detect–restart–abstain matches a choke-point-aware controller at lower cost, under graders whose limits we report (Section 9). The question is old in another guise: randomised search algorithms face the same choice between continuing a run and restarting it, and restarts

are often optimal there [39, 40]. Language-model decoding is such a randomised search.

Figure 1 summarises the answer in five panels, one per section that follows.

Every claim carries its evidence level (Section 3): PRE (preregistered before any data), EXP (exploratory) or CONF (a preregistered test on new data). Our contributions:

1. **Where failures are decided.** Early warning is a difficulty signal (question and model alone: AUROC 0.588; the first quarter of the reasoning adds +0.005, n.s., and is at chance within a problem). From 125,832 on-policy continuations with replicated value curves ($r = 0.985$), 33–66% of failures are decided before the first step, and the labelled error is the choke point in 0.438–0.647 of the rest against a null of 0.16–0.20 (PRE; CONF on Llama-3.1-8B, 0.633).
2. **ChokeBench v1**, a benchmark and evaluation protocol (230 failed traces, four generators). Step localisers graded on the human first error *and* the counterfactual choke point. The best PRM rule finds 73% of human first errors but 45% of choke points; a localiser trained and tuned for the choke point does no better (0.39). The benchmark has predictive validity on held-out halves of the rollouts and held-out models: within a trace, naming the choke point raises the recovery gain by +0.40, and the human label adds +0.01 once the choke point is accounted for (EXP).
3. **Why surgery does not pay, proved and measured.** Restarting recovers 44–66% of failed solutions at four attempts and 56–77% at eight. At matched compute, PRM-guided backtracking loses to restart (-0.063, CONF); a free choke-point oracle gains +0.028; and three short propositions explain the pattern: a failed prefix is worth no more than a fresh start in expectation, its only advantage is a bounded token saving, and a restart loop's quality is capped by its detector. In RAG, choosing where to resample adds -0.000 (PRE), and hallucinations do not snowball.
4. **Secondary: a cheap recipe for RAG, with its law.** Detect, restart, abstain (DRA). A preregistered detect–resample controller reduces judged hallucination relative to the base model (-0.047, $p = 0.0003$; the judge's agreement with people is $\kappa = 0.219$, and no human rating of outputs exists yet); plain DRA, tested against the base model on the same items, gives -0.042 [-0.063, -0.022] (EXP), DRA is equivalent to it (H1), at 1.36× the base cost (claim repair 1.71×, the choke-point-aware controller CARE 1.77×, Chain-of-Verification 3.75×). We prove that DRA's residual hallucination rate is the detector's $P(\text{hallucinated} \mid \text{pass})$ whatever the retry budget, which predicts 50.2% → 26.6% [21.0, 32.2] with human labels, and gives an operating curve down to 12.7% at 4.4 generations per answer; a four-fold retry budget raises coverage by +2.4 points while delivered hallucination stays within +1.5 points, as predicted (EXP).
5. **A measurement caution** (EXP). The gain of a detector-gated loop looks +0.326 under a grader trained like the detector and +0.017 (n.s.) under an independent judge: the gate partly grades itself.

Scope of the claims. Proposition 1 is conditional: it bounds the per-attempt value of a resampling point chosen without lookahead, on failed traces sampled from the generator itself. It does not say restart is always optimal. Rules that look ahead, or shorter attempts (Proposition 2), can win, and the ProcessBench traces of Sections 5–7 are only near on-policy. Empirically, restart beat PRM-guided backtracking on the held-out Llama-3.1-8B (CONF), but restart against backtracking to the human error was not confirmed (C1: -0.022 [-0.090, +0.047]). On exactly on-policy traces of a reasoning model (Section 8), Proposition 1's prediction holds, but a matched-budget gain through shorter attempts could not be ruled out.

All code, rollouts, tables and the generator of every number in this paper are released.[1]

2 Related Work

Process supervision. Step-level supervision outperforms outcome supervision [2, 3]; Monte Carlo rollouts can label steps automatically [4], but such labels disagree with human judgements because a completion can reach the right answer from a wrong step and vice versa, which Zhang et al. [6] treat as label noise. ProcessBench [5] evaluates first-error identification. We study the disagreement itself.

Predicting failure early. Models carry information about their own correctness [22, 21], and hidden states predict CoT success before or during generation [9, 10]; conformal methods certify reasoning prefixes [20]. Bulut [11] shows that such pooled AUROCs are largely explained by problem difficulty and fall to chance within a problem. We apply the same control to text detectors on human-annotated traces.

Which steps matter. Resampling from prefixes identifies influential sentences [12], tokens [13, 14] and the importance of the first step [15]. Merrill and Srivastava [16] localise the point of no return for deceptive behaviour and name incorrect solutions as future work; recoverability of error states has been studied for distillation [17]. None compares a counterfactual decision point with human first-error labels. Work on performative CoT [18, 19] finds that answers are often fixed apart from the written reasoning, which our non-fatal errors echo.

Correcting and backtracking. Unaided self-correction rarely helps [8], but models can correct a mistake once told where it is [7], which motivates verifier-guided search that returns to the first error. Independent resampling from scratch, as in self-consistency [23], is the natural baseline for any such scheme; we compare the two directly. Compute-matched comparisons of verifier-guided search with independent sampling are known to depend on difficulty and budget [29], and trained backtracking beats parallel best-of-n on deep search trees but not on shallow ones [36]. Bulut [11] compares truncated reasoning prefixes with matched-token restarts for two reasoning models and finds that long prefixes mostly buy the same successes more cheaply; we study *failed* traces, with expert error labels, four generators and verifier costs charged. Persistent-pool PRM search revisits discarded prefixes [37]; our results bound what any such revisiting can gain when the

[1] https://github.com/HarshDubey23/early-failure-detection-cot

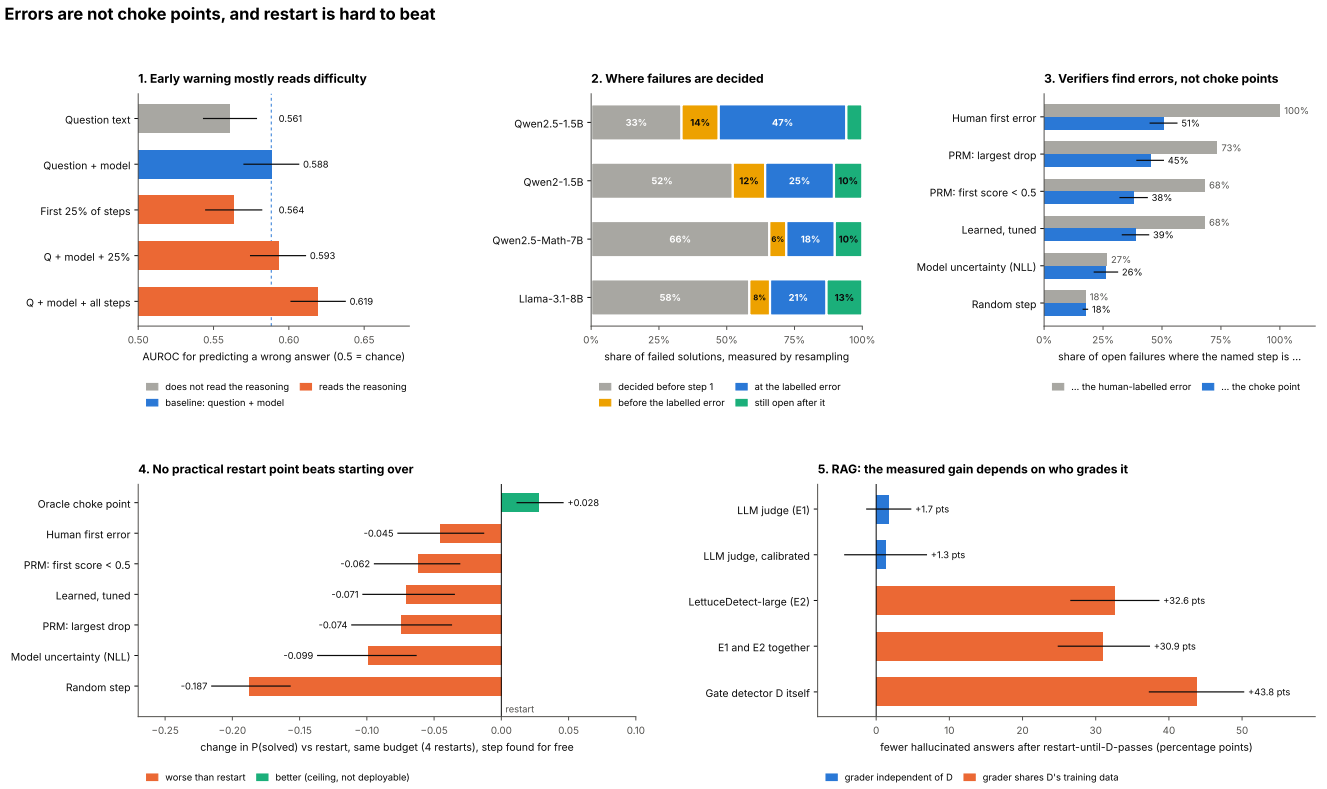


Figure 1: Why surgical repair does not pay. (1) Early-warning detectors mostly read difficulty (Section 4). (2) Where failures of four generators are decided, from resampling (Section 5). (3) Step localisers find human-labelled errors, not choke points (Section 7). (4) No deployable resampling point beats restart at a matched budget, even with localisation free (Section 7). (5) In RAG, the measured gain of a detector-gated loop depends on whether the grader shares the detector's training data (Section 9).

prefix belongs to a failed on-policy trace. Restart strategies for randomised search have a long theory [39, 40]; Appendix A gives the analogue for language-model decoding.

Hallucination recovery. Detect-then-revise systems rewrite unsupported claims [33] or verify and regenerate [32]; token-level detectors such as LettuceDetect [31], trained on RAGTruth [30], make the gate cheap. Hallucinations can snowball when a model commits to an early claim [35]; we test for this with human span labels and find no excess hazard. Resampling-based interpretability measures the importance of reasoning steps counterfactually [12, 38]; we apply the same tool to human error labels, verifiers and recovery policies.

What is and is not new. Counterfactual resampling from prefixes [12, 38, 16] and matched-budget comparisons of prefixes with restarts [11] already exist. Our contribution is narrower: a comparison of counterfactual decision points with expert first-error labels, on failed traces of several generators, together with compute-matched tests of the recovery policies those labels motivate. ChokeBench is an initial resource built from these experiments; whether its mismatch holds for new models and datasets, and whether it predicts which interventions work, is for others to test.

3 Data and Definitions

Data. ProcessBench [5] contains 3,400 step-segmented solutions to GSM8K, MATH, OlympiadBench and Omni-MATH problems (2,842 distinct problems) from 12 open models, each with the index e of the first erroneous step (or none) and whether the final answer is correct. Correct and incorrect answers are balanced within each tier by construction.

Outcome groups. A trace is *clean* (no annotated error; 1,179), *recovered* (an error, but a correct answer; 521) or *fatal* (an error and a wrong answer; 1,700). No clean trace has a wrong answer. The *fatality* of first errors is $P(\text{wrong} \mid \text{error})$.

Counterfactual value. For a trace $T = (s_1, \ldots, s_n)$ written by model M, let $v_k = P_M(\text{correct} \mid q, s_{1:k})$, estimated by sampling m continuations of the prefix from M; v_0 is M's from-scratch solve rate and v_n is the observed outcome. The *decisive step* is $d = \arg\max_k(v_{k-1} - v_k)$, and the *point of no return* is $c = \min\{k : v_j \leq \delta \; \forall j \geq k\}$. A failing trace is *capability-limited* if $v_0 \leq \delta$, shows *latent commitment* if $c < e$, *commits at the error* if $c = e$, and has a *recoverable error* if $c > e$.

Evidence levels. The hypotheses H1–H4 of the counterfactual experiment (the commitment categories, the decisive step, sample-split backtracking, $\delta = 1/8$ with 1/16 and 1/4 as sensitivity) were committed before any

rollout. Analyses added after seeing the Run A/B data (restart vs. first error, repeated splitting, the position-matched null) are *exploratory*. Before the confirmatory run (Run C) we committed a second plan (git commit `459ff22`; `docs/PREREGISTRATION_RUN_C.md`). It specified:

- tests C1–C4 on Llama-3.1-8B, a new family whose traces cover 479 new questions;
- measurement checks R1–R3 on three new independent sampling batches;
- the practical policies and their cost accounting.

Run C reached 144 of 274 Llama fatal traces within its 11.5-hour limit. Traces were processed in a random order, and recovered and clean traces were scheduled after the fatal ones, so none of those were run. All Run C deviations are reported in the text.

Statistics. Detectors are TF-IDF (uni+bigrams) or one-hot features with logistic regression [28], cross-fitted over 5 folds grouped by problem and repeated over 5 seeds. Intervals are 95% question-level cluster bootstrap intervals [27] (1,000 resamples); differences between detectors use a paired bootstrap on the same traces.

4 Observational Results

4.1 Early failure prediction is a difficulty signal

Table 1 and Fig. 2 compare detectors that read the question, the generator and tier, the reasoning prefix, or combinations. The question alone reaches 0.561 and question plus generator 0.588 [0.570, 0.607]. A detector reading only the first 25% of the reasoning is *worse* than that (-0.025, $p = 0.012$), and adding the 25% prefix to question and generator changes AUROC by +0.005 [-0.004, +0.014]. The reasoning adds +0.022 at 50% and +0.031 [+0.020, +0.041] at 100%, consistently across all five seeds.

Within a problem the generator is the only varying factor besides the reasoning: over 326 problems with both a failing and a succeeding trace (466 pairs), the 25% prefix orders pairs correctly 0.485 of the time [0.440, 0.536] and the full prefix 0.539 [0.487, 0.589], neither better than generator identity alone (0.548). Early failure prediction on this benchmark is therefore almost entirely a statement about which problems are hard for which models.

4.2 One first error in four is not fatal

Fatality of first errors is 0.765 [0.748, 0.782]: 23.5% of annotated first errors are followed by a correct answer (Fig. 3). Fatality falls with difficulty, from 0.966 on GSM8K to 0.659 on Omni-MATH, and is flat in the position of the error (0.749–0.766 across onset quartiles; 0.890 only when the error is in the final step). Fatal and recovered errors occur at the same relative position (Mann–Whitney $p = 0.21$, $r = 0.04$).

Explicit re-checking language after the error is barely more frequent in recovered than in fatal traces (18.6% vs 15.5%, $p = 0.09$), so self-correction does not explain most recoveries. Human coding of a stratified sample of recovered traces by two independent coders is in progress (protocol released); Section 5 measures the same question without coders, by forcing the model to answer at every step.

Table 1: AUROC for a wrong final answer (out-of-fold, 3,400 traces). Q: question text; G: generator and tier.

Detector	AUROC	95% CI	SD (5 seeds)
Q	0.561	0.543–0.578	0.005
G	0.539	0.521–0.557	0.002
Q+G	0.588	0.570–0.607	0.003
Prefix 25%	0.564	0.545–0.582	0.006
Prefix 100%	0.600	0.582–0.619	0.004
Q+G+prefix 25%	0.593	0.575–0.611	0.004
Q+G+prefix 50%	0.610	0.592–0.628	0.004
Q+G+prefix 100%	0.619	0.601–0.637	0.003
Trace length	0.542	0.524–0.561	0.002

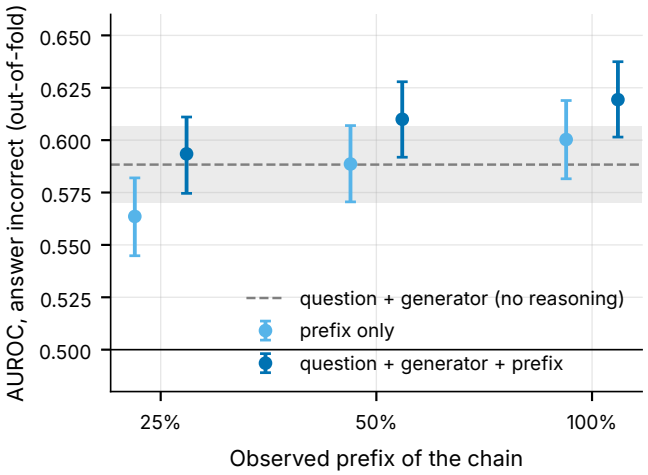


Figure 2: The prefix adds nothing beyond question and generator until half the chain is observed.

4.3 Detectors track errors, not commitment

If a failure detector captured commitment, it would score recovered traces like clean ones. It does the opposite (Table 2): trained to predict a wrong answer, the full-prefix detector separates recovered from clean traces (0.703) but ranks fatal traces *below* recovered ones (0.456 [0.428, 0.484]); with question and generator included the ranking is at chance (0.501). A detector trained to find errors inverts more strongly (0.376). Among erroneous traces, the fatality of the first error is predicted by question and generator (AUROC 0.671), and the error step, the prefix up to it or the next one or two steps change this by at most 0.003.

5 Counterfactual Commitment

The observational results show that first errors and outcomes come apart. They do not show where a failing trajectory actually becomes unrecoverable. We measure that directly by

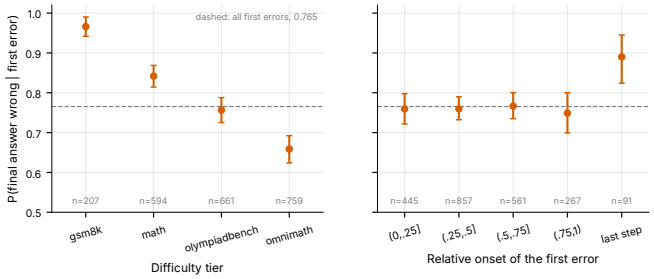


Figure 3: Fatality of first errors by difficulty tier and by where the error occurs.

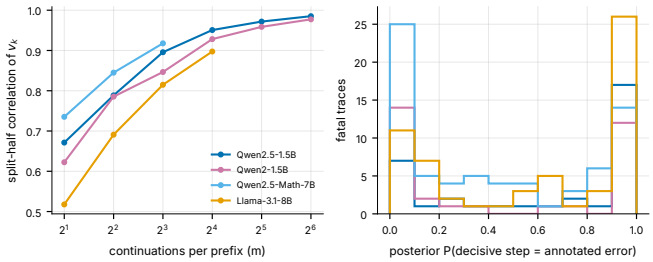


Figure 4: Left: split-half correlation of v_k as a function of the continuations per prefix. Right: posterior probability, per fatal trace, that the decisive step is the annotated first error.

Table 2: AUROC between outcome groups (positive vs negative).

Detector (trained on)	rec. vs clean	fatal vs rec.	fatal vs clean
Prefix 100% (wrong answer)	0.703	0.456	0.664
Q+G+prefix (wrong answer)	0.679	0.501	0.672
Prefix 100% (has error)	0.819	0.376	0.709

resampling.

5.1 Setup

Generators and traces. We use four ProcessBench generators whose weights fit on two 16 GB GPUs, from two model families:

- Qwen2.5-1.5B-Instruct and Qwen2-1.5B-Instruct [24]: all their traces (116 and 88).
- Qwen2.5-Math-7B-Instruct [25]: all fatal and recovered traces with a gold answer, and 60 clean traces (314 in total).
- Llama-3.1-8B-Instruct: the fatal traces completed within the time limit of the confirmatory run (144; Section 3).

The Llama traces cover 512 questions, 479 of which appear in none of the Qwen traces (1,069 distinct questions in total). Gold answers come from the source datasets, matched on problem text, and are compared with `math-verify`.

Rollouts. For every prefix $s_{1:k}$, $k = 0, \ldots, n-1$, we sample m continuations from the trace's own generator with vLLM [26] in float16, at temperature 0.7 and top-p 0.95, and estimate v_k as the fraction that reach the gold answer. The sample sizes are:

- the 1.5B models: $m = 64$, as four independent sampling batches of 16 with different seeds;
- Qwen2.5-Math-7B: $m = 12$;
- Llama: $m = 16$.

Continuations are capped at 1,536 new tokens for the 1.5B models and at 1,024 for the larger models. At the cap, 17% (Qwen2.5-Math-7B) and 29% (Llama) of continuations end without an answer. In total we score 125,832 continuations. Every continuation, its answer and its correctness are released.

What can and cannot be read off v_k. Under the generating policy, v_k is a martingale. Curves conditioned on the eventual outcome therefore drift even when no single step matters: for any fixed k (and any stopping time),

$$\mathbb{E}[v_k \mid \text{fail}] = \frac{\mathbb{E}[v_k(1-v_k)]}{P(\text{fail})} \le v_0, \qquad (1)$$

because $x(1-x)$ is concave (Appendix A). We therefore never interpret a downward drift of failing curves by itself. We ask whether the drop is *concentrated* at the annotated error, against a position-matched null and the noise floor of clean traces. We evaluate backtracking targets with sample splitting.

5.2 How precise are the value curves?

Each v_k is a binomial estimate, and the decisive step $d = \arg\max_k(v_{k-1} - v_k)$ is a noisy argmax. We quantify both kinds of uncertainty (Fig. 4).

Value curves. With the four independent batches of the 1.5B models split two against two, v_k correlates at 0.985 and 0.976 across all prefixes (Spearman–Brown projection to $m = 64$: 0.993 and 0.988). The random split-half correlation is 0.951 for Qwen2.5-Math-7B ($m = 12$) and 0.896 for Llama ($m = 16$).

Decisive steps. We draw every v_k from its Jeffreys posterior and recompute d 2,000 times. The modal step has posterior probability at least 0.5 in 94%, 100%, 87% and 88% of the not-capability-limited fatal traces. Independent halves of the continuations pick the same decisive step in 0.765, 0.906, 0.648 and 0.533 of traces, against position-matched nulls of 0.156–0.284. The posterior probability that the decisive step is the annotated error is bimodal: for most traces it is either clearly yes or clearly no. Disagreement between decisive steps and labels is therefore not measurement noise.

5.3 Where failing trajectories commit

Table 3 classifies every fatal trace at $\delta = 1/8$.

Many failures are decided before the first step. 33%, 52%, 66% and 58% of fatal traces are capability-limited: the model solves the problem from scratch at most one time in eight, so no step of the trace is a choke point. This is the counterfactual counterpart of Section 4's difficulty result.

When the outcome is still open, the annotated error is often where it closes. Among fatal traces that are not capability-limited, the first error is the largest drop in v in 0.647, 0.438, 0.423 and 0.633 of traces. The position-matched nulls are 0.166, 0.204, 0.159 and 0.187. The differences are +0.481 [+0.330, +0.627], +0.234 [+0.063, +0.407], +0.264 [+0.154, +0.382] and +0.447 [+0.328, +0.569]. For Llama

Table 3: Where fatal traces commit ($\delta = 1/8$). "Decisive": share of not-capability-limited fatal traces whose largest drop in v is at the annotated first error; null: position-matched. Llama: fatal traces only.

	Qwen2.5-1.5B	Qwen2-1.5B	Qwen2.5-Math-7B	Llama-3.1-8B
Traces (clean/fatal/rec.)	53/51/12	15/67/6	60/207/47	–/144/–
Continuations per prefix	64	64	12	16
Capability-limited	0.333	0.522	0.657	0.583
Latent commitment ($c < e$)	0.137	0.119	0.063	0.076
Commits at error ($c = e$)	0.471	0.254	0.179	0.208
Recoverable error ($c > e$)	0.059	0.104	0.101	0.132
Decisive = first error	0.647	0.438	0.423	0.633
position-matched null	0.166	0.204	0.159	0.187
n	34	32	71	60
Drop at error: fatal	+0.251	+0.162	+0.109	+0.179
Drop at error: recovered	-0.129	+0.211	-0.215	–
Noise floor (clean)	+0.058	+0.201	+0.078	–

this was preregistered (C3) and is confirmed. The remaining traces show one of two patterns:

- *latent commitment*: success has already collapsed before the written error (21%, 25%, 18% and 18% of not-capability-limited fatal traces);
- *recoverable errors*: success stays above δ past the error (9%, 22%, 30% and 32%).

Event study. Aligned at the annotated error (Fig. 5), fatal traces show a step change from $r = -1$ to $r = 0$: 0.280→0.029, 0.208→0.046, 0.175→0.066 and 0.241→0.062. Clean traces aligned at a random step drift smoothly upward, as the success-conditioned analogue of Eq. 1 predicts. For recovered traces of Qwen2.5-Math-7B, success *rises* across the annotated error (0.374→0.589), by far more than the drift of clean traces.

Does the model already know its answer? (exploratory) At every prefix we force an immediate answer (65,976 forced answers), giving an *answer curve* a_k. The probe is valid at the endpoints. After the full trace, the forced answer reproduces the trace's own answer in 79–91% of cases for three models, but in only 47% for Qwen2-1.5B, for which the probe is unreliable. From the bare question it is right in 11–14% of cases. Answers become accessible late, mostly after the error position (Fig. 6). In fatal traces, the model's accessible answer was correct just before the error and wrong just after it in only 0.058 (Qwen2.5-Math-7B) and 0.107 (Llama) of traces. In most fatal traces the forced answer was never right (0.850 and 0.824). In recovered traces, the correct answer is already accessible right after the error in 0.340 ($n = 47$) and 0.268 ($n = 97$) of traces. This is a lower bound on recoveries in which the written error did not matter, because answers are computed late even in clean traces.

5.4 Where to backtrack, per attempt

Verifier-guided search resamples a failing trace from some earlier point. We compare four targets on fatal traces:

- restarting from scratch (v_0);
- resampling from just before the annotated first error (v_{e-1});
- resampling from just before the decisive step (v_{d-1});
- the best prefix in hindsight (an in-sample upper bound).

Choosing a step and scoring it on the same continuations would reward noise, so we split every prefix's continuations at random into halves A and B. Half A selects the trace ($v_0^A > \delta$) and its decisive step, and half B scores every target. We average over 50 splits and bootstrap over traces. On a planted-effect simulation in which the first error is the decisive step by construction, this estimator returns $+0.001$, so it does not create a gap.

For the three Qwen models, backtracking to the annotated first error is worse than starting over. This was found exploratorily in Run B. Restart beats first-error backtracking by +0.123, +0.165 and +0.068 (Table 4). On the 48 new continuations per prefix alone, which were sampled after the effect was found, it replicates for both 1.5B models. The pooled difference is +0.140 [+0.078, +0.202] (preregistered R1, confirmed). The steps before the error already lower the chance of success. Resampling from just before the decisive step recovers the loss: it beats first-error backtracking by +0.103, +0.147 and +0.108 (all CIs above zero).

For Llama-3.1-8B, it is not. The preregistered confirmatory tests on the new model family fail:

- restart − first error is -0.022 [-0.090, +0.047] (C1, Holm-adjusted p = 1.00);
- decisive step − first error is +0.010 [-0.030, +0.058] (C2, Holm-adjusted p = 1.00).

This is not an artefact of truncation. With continuations that hit the token limit treated as missing rather than wrong, restart − first error is -0.007 [-0.068, +0.048]. The Llama result is consistent with its localization result: the annotated error is the decisive step more often (0.633), so the step before the error is usually still a good place to resume. Backtracking to human first errors is therefore model-dependent. It is worse than restarting for three Qwen models, but not in general.

Decisive-step backtracking and restart: no difference detected, equivalence mostly untested. Decisive − restart is within ± 0.04 for every model, and every 95% CI includes zero. Equivalence at a margin of ± 0.05 (TOST, 90% CI) holds for Qwen2.5-1.5B and for the pooled new batches. It is inconclusive for the other models, including the preregistered C4 test on Llama. We therefore do not claim that decisive-step backtracking matches restart; we claim only that no difference was detected. It uses 15–29% fewer continuation tokens per attempt. The best prefix in hindsight reaches 0.549–0.678.

Reading. Eq. 1 explains why restart is a strong baseline: for a backtracking point at a fixed depth or at a stopping time of the trace, failed on-policy prefixes are on average no better than the empty prefix. The point just before the first error is neither, because it depends on the next step. So the inequality allows first-error backtracking to beat restart or to lose to it. The data show both outcomes, depending on the model.

6 Practical Policies at a Matched Budget

Section 5.4 compares targets per attempt, using oracle knowledge of the decisive step. A deployed system has to choose a restart point without rollouts and pays for every token. We

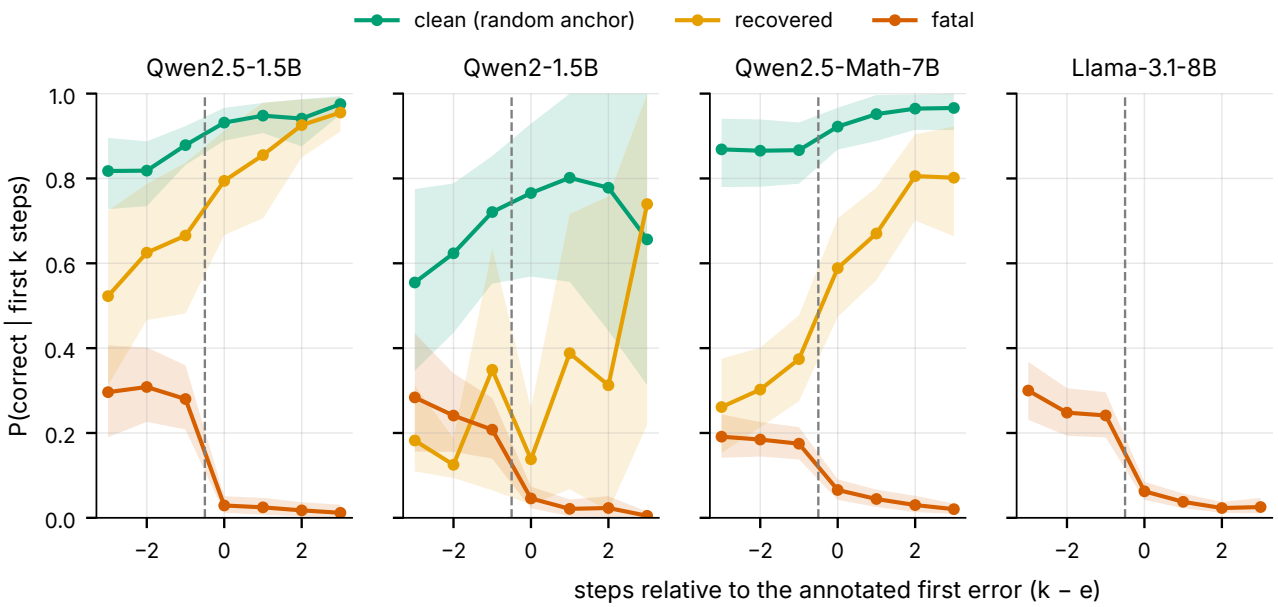


Figure 5: Success probability v_k aligned at the annotated first error ($k - e$; the dashed line separates the last correct step from the error). Shaded: 95% bootstrap CI. Clean traces are aligned at a uniformly drawn step (placebo). Llama: fatal traces only.

Table 4: Success after resampling a fatal trace from each target, per attempt (sample-split, 50 splits; mean pool size in parentheses). 95% trace-bootstrap CIs. Evidence level per row: see Section 3.

	Qwen2.5-1.5B (34)	Qwen2-1.5B (32)	Qwen2.5-Math-7B (76)	Llama-3.1-8B (56)
Restart	0.501	0.575	0.470	0.537
Before annotated first error	0.378	0.410	0.402	0.559
Before decisive step	0.481	0.557	0.511	0.569
Best prefix (oracle, in-sample)	0.549	0.621	0.622	0.678
Restart − first error	+0.123 [+0.054, +0.197]	+0.165 [+0.067, +0.265]	+0.068 [+0.007, +0.133]	-0.022 [-0.090, +0.047]
new batches only (replication)	+0.128 [+0.055, +0.206]	+0.151 [+0.054, +0.249]	–	–
truncated continuations dropped	–	–	+0.107 [+0.044, +0.176]	-0.007 [-0.068, +0.048]
Decisive − first error	+0.103 [+0.053, +0.161]	+0.147 [+0.061, +0.236]	+0.108 [+0.051, +0.165]	+0.010 [-0.030, +0.058]
Decisive − restart	-0.020 [-0.052, +0.013]	-0.018 [-0.061, +0.032]	+0.040 [-0.008, +0.089]	+0.032 [-0.017, +0.085]
TOST at ±0.05 (90% CI)	equivalent	inconclusive	inconclusive	inconclusive
Continuation tokens: restart / decisive	376 / 289	210 / 178	650 / 463	430 / 316

therefore compare practical policies against restart under a matched end-to-end budget.

Policies. All were fixed before the confirmatory data existed (Section 3).

- Restart.
- PRM first error: resample from before the first step that Qwen2.5-Math-PRM-7B [6] scores below 0.5.
- PRM largest drop: resample from before the step with the largest fall in PRM score.
- Probe search: two probe continuations at each of three prefixes, then resample from the latest prefix with a success.
- Learned: logistic regression on step features (PRM score and drop, the generator's NLL and entropy, position, length), trained to predict the decisive step on the *other* generators. For Llama, it is trained only on the three Qwen models.
- Two references: human first error (the expert label, not available in deployment) and the oracle decisive step (all its rollouts counted).

Implementation check. Our implementation of the verifier loads the reward head directly from the checkpoint. It reaches a ProcessBench F1 of 73.2 over 3,374 traces (GSM8K 81.8, MATH 77.4, OlympiadBench 67.0, Omni-MATH 66.7). The published value is 73.5 [6].

Cost.

- Tokens are weighted by model size relative to the generator (FLOPs $\propto$ parameters $\times$ tokens).
- Every prefill (question plus prefix) counts once per restart point, assuming prefix caching.
- Every generated token counts.
- Verifier passes, probes and oracle rollouts are charged to the policies that use them.
- The budget is b times the cost of one restart attempt, for each trace.
- A policy spends what is left on independent attempts from

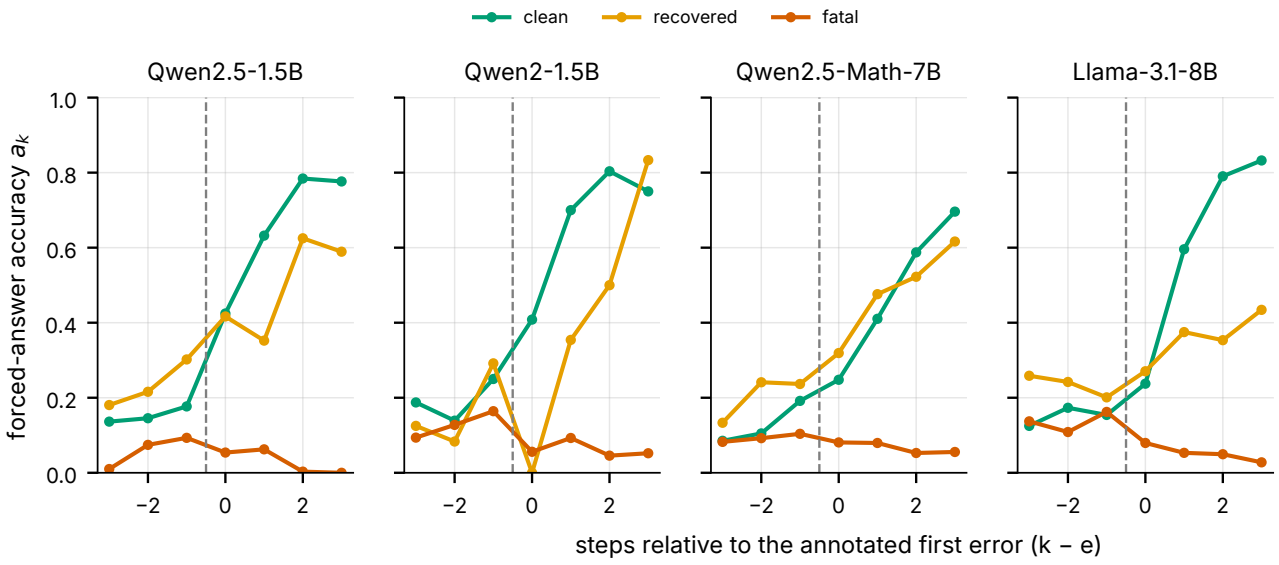


Figure 6: Answer curves: accuracy of an immediately forced answer after k steps, aligned at the annotated first error (clean traces at their middle step).

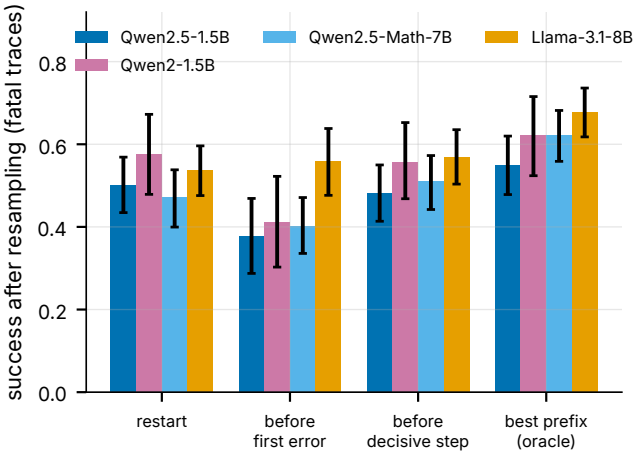


Figure 7: Success after resampling fatal traces from each target (half-B estimates, 50 splits).

its restart point. We report the probability that at least one attempt is correct (Jeffreys-posterior estimate from held-out continuations, 50 splits, trace bootstrap).

No practical policy beats restart reliably (Table 5, Fig. 8).

- **Llama (confirmatory).** PRM largest-drop backtracking is *worse* than restart at four attempts, -0.063 [-0.100, -0.024]. So is the learned policy transferred from Qwen, at -0.187 [-0.226, -0.148].
- **The 1.5B models.** Scoring the trace with a 7B verifier costs about five generator attempts, so every verifier-based policy loses heavily at small budgets.
- **Qwen2.5-Math-7B.** PRM largest-drop backtracking is the only practical policy with a positive estimate. Under full accounting it is -0.022 [-0.052, +0.010] at four attempts and +0.030 [+0.003, +0.057] at eight. If the verifier pass is assumed to be paid anyway, it is +0.046 [+0.017, +0.075] (exploratory).
- **Probe search and the oracle.** Both spend most of the budget on finding the restart point, and lose.

Even a perfect first-error verifier is no shortcut. The human first error, which no deployed system has, is within a few points of restart at matched compute for every model: -0.047, -0.075, +0.039 and +0.001. Later restart points buy more attempts, but each attempt is worth less, which is the trade-off the proposition in Appendix A anticipates.

7 ChokeBench: Grading Localisers on the Target That Matters

ProcessBench grades a verifier by whether it finds the human-labelled first error. The value curves make a second target measurable: the *choke point*, the step where v falls most. ChokeBench grades step localisers on both, and on what the named step is worth as a resampling point. It uses the 230 fatal traces whose outcome is still open ($v_0 > \delta$; effective $n = 199$ after sample splitting) and the released step signals; no model is re-run.

Protocol. A localiser names one step j of a failed trace. We report (i) whether j is the human first error; (ii) the posterior probability that j is the choke point (Jeffreys posterior on every v_k, 2,000 draws), so sampling noise is not counted as a miss; (iii) the per-attempt success of resampling from just before j, minus restart; and (iv) the same difference in P(solved) at a FLOP-matched budget of four restart attempts, *charging nothing for finding* j. Item (iv) is an upper bound on any verifier that uses the localiser; Table 5 charges the verifier. Half of each prefix's continuations select the open traces (and the oracle's choke point), the other half score them, over 50 splits; the estimator reproduces the per-model values of Table 4 within sampling error.

Table 5: P(solved) minus restart for fatal traces at a budget of four restart attempts (FLOP-matched, all overheads charged). 95% trace-bootstrap CIs. "Shared verifier": the PRM pass is not charged, as if it had been run anyway to detect the failure (exploratory). Llama rows are confirmatory for PRM largest drop and Learned; all other rows are exploratory.

	Qwen2.5-1.5B	Qwen2-1.5B	Qwen2.5-Math-7B	Llama-3.1-8B
Restart: P(solved)	0.657	0.514	0.437	0.492
PRM first error	-0.651 [-0.744, -0.558]	-0.514 [-0.603, -0.424]	-0.041 [-0.066, -0.016]	-0.068 [-0.102, -0.035]
PRM largest drop	-0.651 [-0.744, -0.558]	-0.514 [-0.603, -0.424]	-0.022 [-0.052, +0.010]	-0.063 [-0.100, -0.024]
Probe search	-0.646 [-0.739, -0.554]	-0.448 [-0.537, -0.361]	-0.428 [-0.468, -0.388]	-0.489 [-0.539, -0.436]
Learned	-0.657 [-0.749, -0.563]	-0.514 [-0.603, -0.424]	-0.151 [-0.177, -0.127]	-0.187 [-0.226, -0.148]
Human first error (reference)	-0.047 [-0.122, +0.027]	-0.075 [-0.132, -0.022]	+0.039 [+0.012, +0.066]	+0.001 [-0.025, +0.029]
Oracle decisive step, rollouts counted	-0.657	-0.514	-0.437	-0.492
PRM largest drop, shared verifier	-0.061 [-0.146, +0.022]	-0.098 [-0.160, -0.040]	+0.046 [+0.017, +0.075]	+0.003 [-0.030, +0.036]

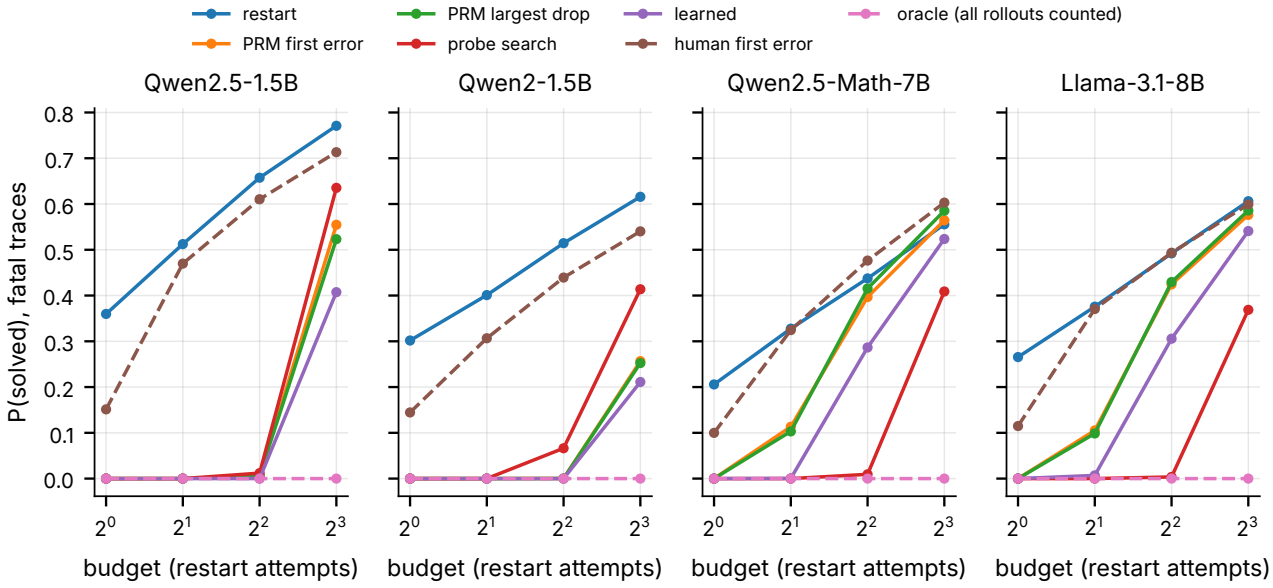


Figure 8: P(solved) for fatal traces against the end-to-end budget (in restart attempts, FLOP-matched, all overheads charged).

Localisers find errors, not choke points (Table 6, Fig. 9). The PRM's largest-drop rule names the human first error in 73% of open failures, consistent with its ProcessBench F1, but the choke point in only 45% [0.39, 0.51]. The expert label itself is the choke point with probability 0.51. Generator uncertainty (maximum token NLL or entropy) is far below the PRM on both targets, though above a random step. The beginning of the trace is the choke point in 28% of open failures: many of them go wrong on the first step.

Even a perfect localiser buys little. Per attempt, every deployable localiser is worse than restart, from -0.077 to -0.178, and so is the human label (-0.066). With the step found for free and the shorter continuations converted into extra attempts, the human label is still -0.045 [-0.077, -0.013] and the best PRM rule -0.062. An oracle that knows the choke point from held-out rollouts reaches +0.028 [+0.012, +0.046]. That is the ceiling for any backtracking verifier on these traces, before its own cost is paid. It is small because, before the choke point, a failed trace is worth about as much as a fresh start (Appendix A): finding the choke point saves tokens, not successes.

Can a localiser be learned? We train one directly for the choke-point target. Each step of a failed trace becomes an example described by nine features (PRM score, PRM drop, first PRM score below 0.5, mean and maximum token NLL and entropy, step length, relative position), labelled softly with the posterior probability that it is the choke point. The model names the step with the highest predicted probability. Evaluation is leave-one-generator-out, with nested selection: for each held-out generator, 24 configurations (logistic regression with $C \in \{0.01, 0.1, 1, 10\}$ and gradient-boosted trees with depth $\in \{2, 3\}$ and learning rate $\in \{0.05, 0.2\}$, each on PRM, generator or all features) are scored by leave-one-generator-out cross-validation on the other three generators only, and the best is refit and tested once (Fig. 10). The tuned localiser finds the choke point with probability 0.39 [0.33, 0.44], no better than the fixed PRM rules, and its restart point is worth -0.071 at a 4× budget. Inner-CV scores do not carry over to the held-out model (best inner score 0.51 vs test 0.25 for Qwen2.5-Math-7B): which step decides a failure is model-specific, as

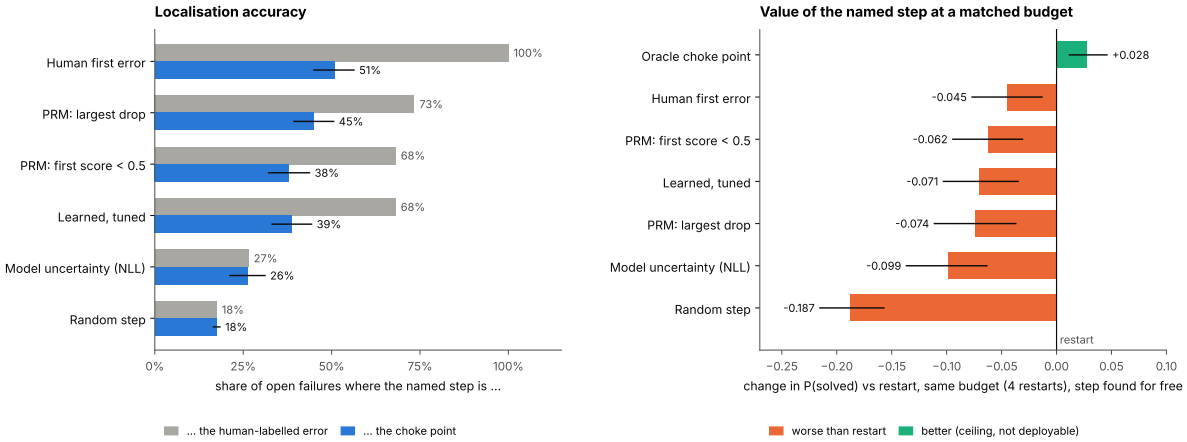


Figure 9: ChokeBench. Left: PRM rules find human-labelled errors far more often than choke points. Right: P(solved) minus restart at a budget of four restart attempts when finding the step is free.

Table 6: ChokeBench, pooled over four generators. Accuracy on the human first error (Human) and on the choke point (Choke, posterior); success minus restart per attempt and at a 4× budget with free localisation. 95% trace-bootstrap CIs in Fig. 9.

Localiser	Human	Choke	Δ/attempt	Δ at 4×
Human first error	1.00	0.51	-0.066	-0.045
PRM: first step < 0.5	0.68	0.38	-0.077	-0.062
PRM: largest drop	0.73	0.45	-0.103	-0.074
PRM: minimum	0.61	0.39	-0.158	-0.121
Generator: max token NLL	0.27	0.26	-0.097	-0.099
Generator: max entropy	0.30	0.28	-0.178	-0.165
Random step	0.18	0.18	-0.224	-0.187
First step (= restart)	0.14	0.28	+0.000	+0.000
Learned, tuned (held-out model)	0.68	0.39	-0.085	-0.071
Oracle choke point (split)	–	–	+0.015	+0.028

the transfer failure of the learned restart policy in Section 6 already suggested.

Does a ChokeBench score predict recovery? (EXP.) A benchmark for localisers is useful only if scoring well on it predicts that resampling from the named step helps. We test this without circularity. In each of 50 random splits of every prefix's continuations, half A labels the choke point and half B scores the gain of resampling just before the named step; the halves are independent given the true value curve. Over 232 open failed traces and every localiser, with trace fixed effects (so problem difficulty cancels), naming the choke point raises the per-attempt gain by +0.40 [+0.37, +0.44], and naming the human first error by +0.15 [+0.11, +0.20]. Fitted jointly, the human label adds nothing once the choke point is accounted for (+0.01 [-0.03, +0.05], against +0.40 for the choke point). The human label is useful only to the extent that it coincides with the choke point. The coefficients compare named steps within a trace, not a named step with restart: hitting the choke point avoids most of the loss that other steps incur, which is why even the oracle only ties restart. At the system level, ranking localisers by their ChokeBench score on three generators predicts their mean gain on the fourth better than ranking them by a ProcessBench-style score: mean Spearman $\rho = +0.52$ against +0.22, difference +0.29 [+0.08, +0.46] (trace bootstrap; seven localisers per generator, so each single ρ is noisy). Localiser-level accuracies and gains had been seen before this analysis, so it is exploratory (`scripts/54_chokebench_validity.py`).

Release. `results/chokebench/chokebench_v1.jsonl` holds, for every failed trace, the per-prefix success counts, continuation counts and lengths, the human label and the PRM scores. A new localiser is graded by naming one step per trace (`scripts/49_chokebench.py`).

8 Reasoning Models, Exactly On-Policy (Run D)

The ProcessBench traces above are near on-policy, and their generators are instruction models. Run D (preregistered, `docs/PREREGISTRATION_RUN_D.md`, with deviations recorded before launch) asks the same questions of two long-reasoning models whose traces are exactly on-policy: Qwen3-1.7B in thinking mode and DeepSeek-R1-Distill-Qwen-1.5B, on MATH-500 problems 21–500, with a 6,144-token budget per answer. Each model writes one trace per problem. Each failed trace is cut at fixed token positions $t \in \{0, 256, \ldots, 5120\}$, moved back to the last paragraph break so that a cut depends only on the past, and $m = 8$ continuations are sampled from each cut under the same total budget (Fig. 11).

Failures are mostly budget exhaustion. Accuracy is 70.8% (Qwen3) and 72.7% (R1-distill). 95% and 92% of their failures end because the reasoning hits the budget, not because of a wrong final answer: these failures have no wrong step for a localiser to find.

Value curves. On 2×T4 GPUs only R1-distill's stage 2 produced data in the session limit: the first 40 failures in dataset order. Proposition 1's prediction (D1) holds: no cut

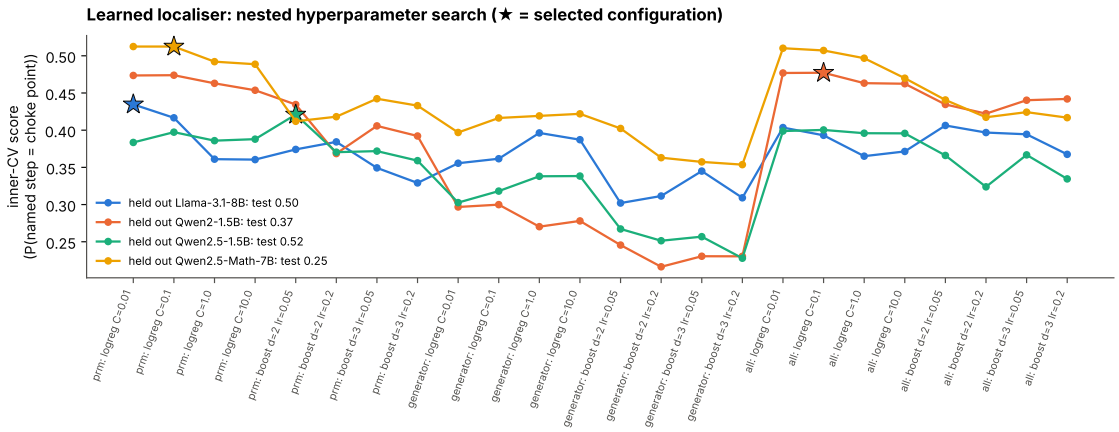


Figure 10: Nested hyperparameter selection of the learned localiser. Each line is one outer fold (held-out generator); points are inner leave-one-generator-out scores of the 24 configurations; stars mark the selected configuration; the legend gives its score on the held-out generator.

is worth more than a fresh start, and $v(t) - v(0)$ falls from 0 at $t = 256$ to -0.212 [-0.303, -0.128] at $t = 5120$. 60% of these failures are capability-limited. At a matched budget of generated tokens (D2), the preregistered estimator finds the best cut ahead of restart by +0.133 [+0.067, +0.197] at 4×, so the preregistered "restart not beaten" criterion fails. The gain is fragile. Per attempt no cut beats restart, so it comes entirely from cheaper attempts: a late cut leaves a short budget, and its attempts cost a fifth of a restart's. The estimator extrapolates four observed samples to up to about 19 attempts, and its Jeffreys prior credits a cut with no observed success with about 10% per attempt. A plug-in estimate gives +0.031 [-0.045, +0.114] (exploratory). On reasoning models, resampling late in a failed trace may therefore pay through shorter attempts, as Proposition 2 allows, but our samples cannot settle it. Budget-truncated traces also make part of the fall in $v(t)$ mechanical, since later cuts leave less budget; only 4 wrong-answer failures have value curves (D4).

9 The Same Question for Hallucinations

Retrieval-augmented generation (RAG) offers what mathematics lacks: a detector can check, while the system runs, whether each sentence is supported by the source. If hallucinations were choke points, a controller that estimates v_k online and resamples from the most recoverable boundary should beat plain restarting. We built such a controller and preregistered its test (`docs/PREREGISTRATION_RECOVERY.md`, committed before any data).

Setup. RAGTruth [30] test sources (450; QA, summarisation, data-to-text), whose Llama-2-7B-chat responses carry human span labels, and 300 HotpotQA [34] items with controlled evidence (half with the answer removed). The gate D is LettuceDetect-base [31]. When D flags a sentence, the controller (CARE) tries claim repair, then resamples, choosing between restarting and backtracking to just before the flagged sentence by Thompson sampling on each boundary's estimated recoverability per token, then abstains (QA) or deletes what is still flagged. Every policy replays the same bank of on-policy continuations (9,696 in the mechanism study alone), under a budget cap of $b = 2$ draft-equivalents; tokens actually spent are reported. The preregistered primary evaluator E_1 is an LLM judge from another family (Qwen2.5-7B-Instruct; $\kappa = 0.219$ with RAGTruth's human labels, passing the preregistered bar of 0.20); E_2 (LettuceDetect-large) is secondary because it shares training data with D.

Preregistered results (RAGTruth).

- *H0, sanity:* CARE reduces hallucination relative to the base model, -0.047 [-0.072, -0.025] ($p = 0.0003$).
- *H1, versus restart until D passes:* -0.005 [-0.024, +0.013]; equivalent within ±0.03 (TOST 90% CI [-0.021, +0.010]). Restart spends 1,293 tokens per item against 1,677 for CARE.
- *H5, does the commitment-aware choice add anything?* CARE minus the same controller restricted to restarts: -0.000 [-0.005, +0.005] under E_1 and +0.003 [-0.004, +0.010] under E_2. An oracle that knows each boundary's true recoverability gains -0.015 [-0.027, -0.004]: there was little to choose between.
- *Mechanism, with human span labels:* over 226 human-hallucinated drafts, resampling from just before the first hallucinated sentence is no better than restarting (+0.024 [-0.002, +0.049], $p = 0.066$; +0.093 for summaries, exploratory). Keeping a hallucinated sentence does not raise the hazard of later ones relative to a placebo (-0.006 [-0.029, +0.019]): no snowballing [35].

HotpotQA agrees: H0 holds and no comparison against an alternative strategy is significant.

Who grades the loop matters (exploratory). Detector-gated policies resample until D is satisfied, so an evaluator that resembles D will tend to agree that they succeeded. We score every RAGTruth policy on the 232 sources of a held-out

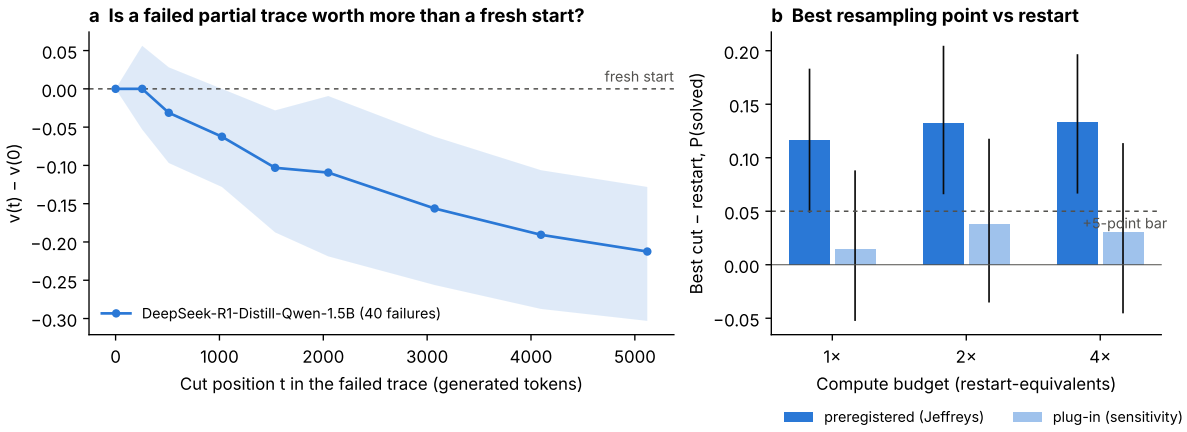


Figure 11: Run D, DeepSeek-R1-Distill-Qwen-1.5B, 40 failed traces. (a) Value of a failed partial trace cut at t tokens, minus a fresh start (95% bootstrap band). (b) Best cut (chosen on half of the continuations, scored on the other) minus restart at a matched budget of generated tokens, under the preregistered Jeffreys estimator and a plug-in sensitivity.

Table 7: Hallucination rate (%) on the held-out half of RAGTruth under five evaluators. D-family evaluators share LettuceDetect's training data with the gate. Differences from restart with 95% CIs: `results/recovery/final/evaluator_robustness.md`.

			independent of D		D-family		
Policy	Cov.	Tokens	E_1	E_1-cal.	E_2	dual	D
Base model	100.0	980	80.7	51.1	47.2	49.8	44.6
Restart until D passes	99.6	1,320	79.0	49.8	14.6	18.9	0.9
Backtrack until D passes	99.6	1,279	80.3	52.8	19.3	26.2	0.4
CARE	97.1	1,708	77.7	46.8	19.3	24.9	2.1
CARE, restarts only	96.8	1,710	77.3	45.1	18.5	23.6	2.1
Abstain if flagged (QA)	88.4	1,001	67.8	42.1	15.0	19.7	5.6
CoVe, revised answer (n=98)	100.0	–	71.4	43.9	41.8	42.9	30.6

half under five evaluators (Table 7). One of them never reads a LettuceDetect score: E_1 with a threshold on the share of sentences it calls unsupported (25%), chosen on the other 216 drafts from a grid fixed in advance. Its human agreement is weak ($\kappa = 0.189$ on the held-out drafts), like E_1's (0.238); E_2, the "dual" rule $E_1 \wedge E_2 \geq 0.3$ and D agree better (0.535, 0.560, 0.476) but share training data with the gate (E_2 agrees with D on 85.0% of drafts, E_1-calibrated on 58.9%). The threshold 0.3 of the dual rule was chosen on all drafts, so its held-out κ is not an out-of-sample estimate.

Two conclusions survive the change of evaluator: CARE beats restart under none of the five, and it matches its restart-only ablation under both preregistered evaluators (H5). Everything else depends on the evaluator. Restarting until D passes lowers the hallucination rate by +0.326 according to E_2 and +0.309 according to the dual rule, but by +0.017 [-0.013, +0.047] according to E_1 and +0.013 [-0.043, +0.069] according to its calibrated version. Chain-of-Verification [32], which never consults D, looks far worse than restart to the D-family (+0.235 under the dual rule) and indistinguishable from it to the independent judges (+0.031 [-0.031, +0.092]). The independent judges are weak, so they cannot rule out a real reduction; but the D-family numbers are partly the gate grading itself. Gains of a verifier loop should be measured by an evaluator that shares no training data with the verifier, and ultimately by people (a blinded two-rater kit is released; Section 12).

Reading. As in mathematics, the step where the output first goes wrong is a fine but not a better place to resume than the beginning, and an online estimate of recoverability does not change that. Hallucinations here behave like local errors rather than commitments: detecting and resampling them reduces hallucination by an amount the evaluators disagree on, and choosing where to resample adds nothing.

10 What Mitigates Failures, and at What Cost

The previous sections asked where failures are decided. This one collects what the same data say about mitigating them cheaply (Fig. 12, Table 8).

Maths: restarting recovers most failures. For the failed ProcessBench solutions of the four generators, plain restarting (independent attempts from the question, FLOP-matched, every token counted) solves 21–36% of them with one attempt, 44–66% within four and 56–77% within eight. No surgical policy in Table 5 does better at any of these budgets, except PRM backtracking for Qwen2.5-Math-7B at eight attempts (+0.030, exploratory).

RAG: detect, restart, abstain. The detector D is a 150M-parameter token classifier, so checking a draft costs little; only flagged drafts are resampled, and what is still flagged after the budget is withheld (QA) or deleted (other tasks). This loop (DRA, "restart until D passes" in Section 9) is the cheapest mitigation that resamples: 1,293 tokens per item against 948 for the base model (1.36×), while claim repair costs 1.71×, the choke-point-aware controller 1.77× and Chain-of-Verification 3.75×. Deleting flagged sentences costs almost nothing (1.02×) but changes the content of the answer; abstaining on flagged QA gives the lowest hallucination under every grader at a coverage of 87.1%.

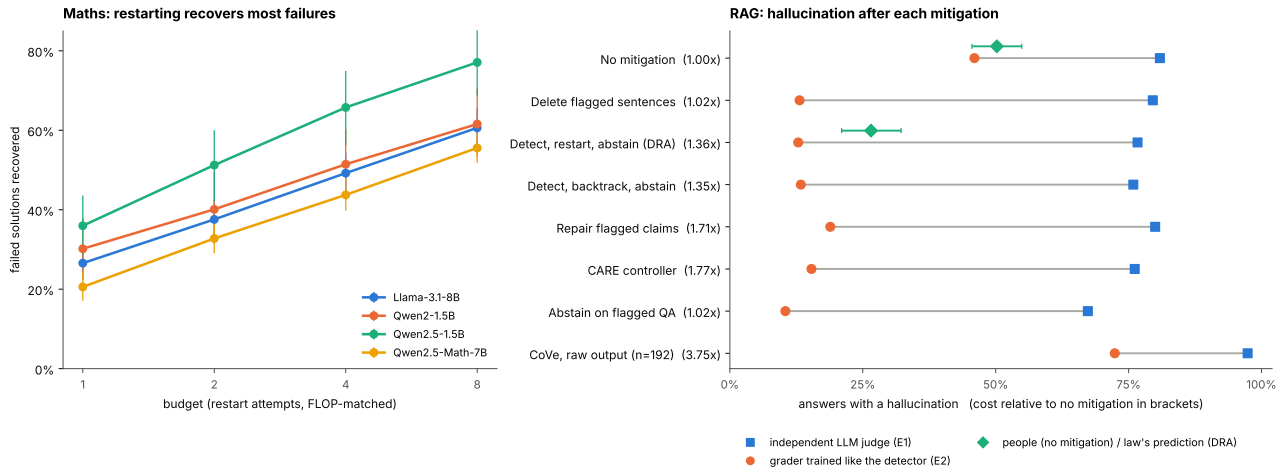


Figure 12: What mitigation buys. Left: share of failed maths solutions solved by plain restarting, by budget (95% CIs). Right: hallucination after each RAG mitigation under the independent judge (E_1) and the detector-family grader (E_2), with the human base rate and the human-scale prediction of Proposition 3 (Appendix D) for DRA; cost relative to no mitigation in brackets.

Table 8: Mitigation on RAGTruth (450 sources, Llama-2-7B-chat). Tokens per item, every call counted; hallucination rate (%) under the independent judge E_1 and the detector-family grader E_2. CoVe: raw outputs on 192 items (the echoed verification block inflates both graders, Section 9).

Mitigation	Tokens	Cost	E_1	E_2	Coverage
None (base model)	948	1.00×	80.9	46.0	100.0
Delete flagged sentences	968	1.02×	79.6	13.1	99.8
Detect, restart, abstain (DRA)	1,293	1.36×	76.7	12.9	98.9
Detect, backtrack, abstain	1,279	1.35×	75.9	13.4	97.1
Claim repair, then delete	1,624	1.71×	80.0	18.9	99.8
CARE (choke-point-aware)	1,677	1.77×	76.2	15.3	96.5
Abstain on flagged QA	968	1.02×	67.3	10.4	87.1
Chain-of-Verification	3,558	3.75×	97.4	72.4	100.0

How much does DRA reduce hallucination for people? The two graders disagree by an order of magnitude, and no human rating of system outputs exists yet. Proposition 3 (Appendix D) gives a principled estimate. If the system returns the first independent draw that D passes, the share of returned answers that are hallucinated is $P(\text{hallucinated} \mid D \text{ passes})$, whatever the retry budget. RAGTruth's human labels on the drafts measure exactly this quantity: D misses 28.4% of hallucinated drafts and flags 20.6% of faithful ones, so the predicted rate among DRA's answers is 26.6% [21.0, 32.2], against 50.2% [45.5, 54.9] for the base model: a reduction of 23.7 points [19.4, 28.2]. Two assumptions carry this estimate (draws independent given the input, and the detector's error rates on resampled answers equal those on drafts), and with per-item variation in hallucination rates it is optimistic, so we read it as an upper bound on the reduction and report it as exploratory. An attempt to confirm it by inverting the independent judge's own error rates was consistent (24.8%) but too imprecise to be informative ([-5.0, 43.5]); the human rating kit is released for that purpose.

Two checks of the law (EXP, Fig. 13). *Budget invariance.* The law predicts that more retries raise coverage but leave the hallucination of delivered answers unchanged, for any fixed definition of "hallucinated". The replay ran DRA at budgets 1 and 4 on the same 450 sources. Coverage rose from 97.2% to 99.6% (+2.4 points [+1.5, +3.4]), while delivered hallucination moved by +0.6 points [-0.1, +1.4] under E_2 and +1.5 [+0.9, +2.2] under E_1. The small rise is the direction the heterogeneity caveat predicts: the extra answers come from inputs that are harder to answer faithfully. *Operating curve.* Sweeping the detector threshold, the law turns the human-labelled drafts into a menu of hallucination rates and their price (Fig. 13, right). With the setting of our runs (D, $\tau = 0.5$) it predicts 21.6% at 2.2 generations per answered input; a gate that flags when either LettuceDetect model fires at $\tau = 0.1$ predicts 12.7% [7.1, 19.1] at 4.4, against 50.2% without mitigation. These predictions inherit the law's assumptions and await human ratings of system outputs.

What the law says about cost. The retry budget only buys coverage: abstentions fall geometrically with more attempts, but the hallucination rate of delivered answers stays at the detector's floor. The lever is therefore the detector, not the budget and not the resampling point: a better detector lowers the floor, and restarting is the cheapest way to cash it in.

11 Discussion

For practitioners: how to mitigate cheaply. Spend compute on a cheap, accurate detector and on restarts, not on finding where the output went wrong. Gate every output with the detector; restart only what it flags; abstain (or delete the flagged sentences) when the budget runs out. Proposition 3 makes the trade-off explicit: the detector sets the hallucination rate of what is delivered, the budget sets how often the system answers, and the resampling point sets neither. Report the gain with a grader that shares no training data with the detector.

For early warning. Reported AUROCs for early failure

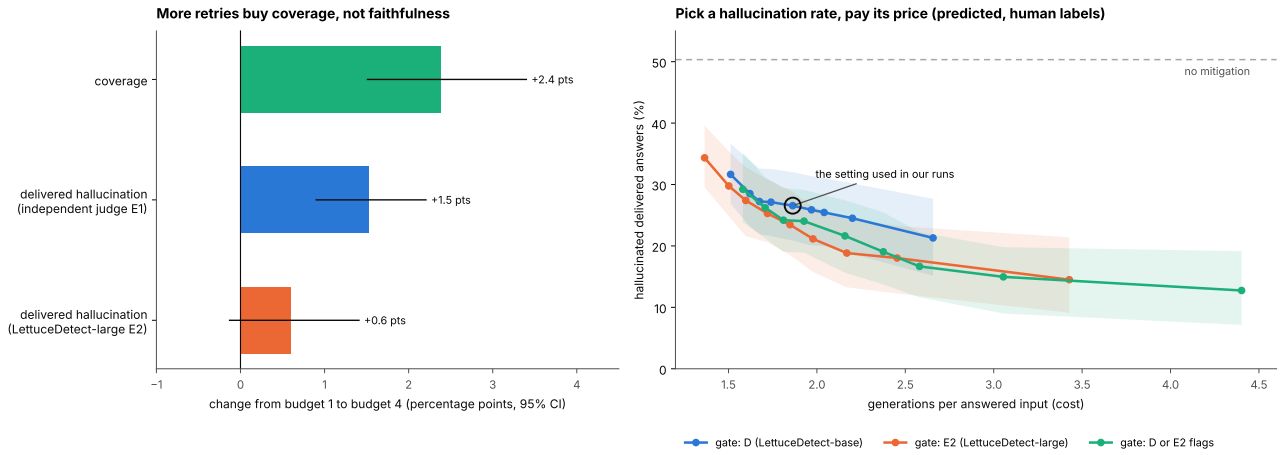


Figure 13: Checks of Proposition 3. Left: raising the DRA budget from 1 to 4 raises coverage while delivered hallucination stays nearly flat (paired over 450 sources, 95% CIs). Right: predicted hallucination of delivered answers against generations per answered input, for three gates and a sweep of thresholds, from 449 human-labelled drafts (bands: 95% bootstrap CIs); the circle marks the setting of our runs.

prediction should come with a question-only baseline, a generator baseline, the increment over both, and a within-problem concordance. Without them, a detector of problem difficulty cannot be told apart from a detector of reasoning failure. On ProcessBench, text detectors pass this test only after half the chain. The counterfactual measurement explains why. Between 33% and 66% of failures are fixed before the first step, so reading the trajectory can add little for them.

For process supervision. Human first-error labels mark where reasoning first goes wrong, which is the right target for auditing. They are also, more often than chance, where success collapses: 0.423–0.647 of open failures, confirmed on a new model family. But using them as restart points depends on the model. Per attempt it was worse than restarting for three Qwen models and equal for Llama, and at matched compute it never clearly beat restarting. Verifier-guided backtracking should therefore always be reported against restart at matched end-to-end compute, with the verifier's own cost included.

What would change the picture. The in-sample oracle shows apparent headroom: the best restart point in hindsight succeeds 0.549–0.678 per attempt, against 0.470–0.575 for restart. ChokeBench shows that most of it is selection on noise: chosen and scored on independent halves of the rollouts, the choke point is worth +0.015 per attempt and +0.028 at a 4× budget, before any verifier cost. Backtracking can win only where a prefix holds work the model cannot reproduce from scratch, which is the regime of long, correct-so-far reasoning [11] and of deep search problems [36], not of failed traces at these scales. The released value curves let anyone test a localiser against that ceiling.

For grounded generation. The math result carries over: the first hallucinated sentence is not a better resampling point than the beginning, an online recoverability estimate does not change that, and hallucinations do not snowball. A detector-gated loop does reduce hallucination, but by how much depends on whether the evaluator resembles the gate. We recommend that recovery systems report (i) a matched-compute restart baseline, (ii) an ablation that removes the choice of resampling point, and (iii) an evaluator that shares no training data with the gate.

12 Limitations and what remains unverified

- **Coverage.**
  - Four open models of at most 8B parameters, mathematics only.
  - Llama contributes 144 fatal traces. Its recovered and clean traces were not run, so Llama has no noise floor, no event study for those groups, and no answer-curve comparison by rollout.
  - Qwen2.5-7B-Instruct and larger models were not run.
- **On-policy approximation.** ProcessBench re-segmented the original solutions, and the decoding settings that produced them are unknown. Prefixes are therefore near-, not exactly, on-policy, and the proposition holds only approximately for them.
- **Truncation.** For the larger models, 17% and 29% of continuations hit the token cap without an answer. Treating them as missing changes no conclusion, but it lowers absolute success rates.
- **Equivalence.** Equivalence of decisive-step backtracking and restart at ± 0.05 is established only for Qwen2.5-1.5B and the pooled new batches. Elsewhere, including the pre-registered Llama test, it is inconclusive.
- **Exploratory results.** The difficulty and fatality analyses, the answer curves, the shared-verifier accounting and all policy results on the Qwen models are exploratory.
- **Human coding.** The coding of recovered traces by two independent human coders has not been done. The protocol and blinded sheets are released, and no coding result is reported.
- **Hallucination recovery.** One generator (Llama-2-7B-chat);

the Llama-3.1-8B arm produced drafts only (GPU quota), so no recovery claim is made for it. Chain-of-Verification ran on 192 of 750 items. The independent judges agree only weakly with people, and no human rating of the system's outputs exists yet; a blinded 80-pair, two-rater kit is released.
- **ChokeBench** is built from the same rollouts as Sections 5–6 and is exploratory; its localisers are fixed rules, not trained models (the learned policy of Section 6 is the trained case).

13 Conclusion

Errors are not choke points. On failed reasoning traces from four generators, the human-labelled first error is a better-than-chance but imperfect locator of the step where success actually collapses; many failures are decided before the first step; reward models track the label rather than the collapse; and even a free oracle for the collapse buys about three points over restart at four restarts' budget, as a supermartingale bound says it should. Restart, which recovers 44–66% of failed solutions within four attempts, is the baseline any localise-and-repair method must beat at matched compute. In a secondary RAG study the same pattern holds under the graders we have, but those graders agree only weakly with people, and human ratings of system outputs are the next test. Whether the gap between errors and choke points holds for new models and datasets is the question ChokeBench is released to answer.

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A Why restarting is a strong baseline

Let a trace $s_1, \ldots, s_n$ be sampled from policy M for question q, let S be the event that its final answer is correct, $\mathcal{F}_k = \sigma(s_{1:k})$, and $v_k = P_M(S \mid q, \mathcal{F}_k)$. Resampling a fresh continuation of $s_{1:k}$ from M succeeds with probability v_k.

Proposition 1. For every stopping time $\tau \leq n$ of $(\mathcal{F}_k)$, $\mathbb{E}[v_\tau \mid \neg S] \leq v_0$ (whenever $v_0 < 1$).

Proof. (v_k) is a bounded martingale and $g(x) = x(1-x)$ is concave, so $g(v_k)$ is a bounded supermartingale; by optional stopping $\mathbb{E}[g(v_\tau)] \leq g(v_0)$. By the tower property, $\mathbb{E}[v_\tau \mathbf{1}_{\neg S}] = \mathbb{E}[v_\tau(1-v_\tau)] = \mathbb{E}[g(v_\tau)]$. Dividing by $P(\neg S) = 1 - v_0$ gives $\mathbb{E}[v_\tau \mid \neg S] \leq v_0(1-v_0)/(1-v_0) = v_0$. □

Consequences. (i) Any backtracking rule that picks its restart point by reading the failed trace only up to that point (a fixed depth, a confidence threshold crossed so far, a verifier that flags step k using $s_{1:k}$ and resamples from k) cannot, per attempt and in expectation over failed on-policy traces, beat restarting from scratch. (ii) Rules that look ahead can: resampling from *before* a flagged step uses step $k+1$, and the decisive step uses rollouts. The proposition therefore sets the bar a backtracking target must clear, and the paper measures how far human first errors and verifier signals clear it. (iii) The proposition is per attempt. Under a fixed compute budget a later restart point buys more attempts, because continuations are shorter, which is why Section 6 compares policies at a matched end-to-end budget. (iv) ProcessBench traces were not necessarily sampled at the temperature used for v_k, so the proposition holds only approximately for them; the empirical comparisons do not rely on it.

B Preregistered tests and outcomes

Test	Hypothesis (data)	Estimate [95% CI]	Outcome
H1	first error is the decisive step more often than chance (Qwen, Run B)	see Table 3	supported (Qwen2-1.5B CI includes null at $m = 16$)
C1	restart > first-error backtracking (Llama)	-0.022 [-0.090, +0.047]	not confirmed
C2	decisive-step > first-error backtracking (Llama)	+0.010 [-0.030, +0.058]	not confirmed
C3	decisive share > position-matched null (Llama)	+0.447 [+0.328, +0.569]	confirmed (Holm-adjusted p <0.002)
C4	decisive-step ≡ restart within ± 0.05 (Llama)	90%: [-0.008, +0.076]	inconclusive
R1	restart > first error on new batches only (Qwen 1.5B, pooled)	+0.140 [+0.078, +0.202]	confirmed
R2	v_k reliable across independent batches	$r = 0.985$, 0.976	confirmed
R3	decisive step stable under posterior noise	modal ≥ 0.5: 94%, 100%	confirmed
P2, P4	practical policies vs restart at $b = 4$ (Llama)	-0.063, -0.187	both worse than restart

C Threshold and definition sensitivity

	Qwen2.5-1.5B	Qwen2-1.5B	Qwen2.5-Math-7B	Llama-3.1-8B
Restart − first error, $\delta = 1/16$	+0.112	+0.155	+0.068	-0.011
Restart − first error, $\delta = 1/8$	+0.123	+0.165	+0.068	-0.022
Restart − first error, $\delta = 1/4$	+0.136	+0.186	+0.116	-0.031
First error = largest drop, $\delta = 1/16$ / $1/8$ / $1/4$	0.561 / 0.647 / 0.633	0.405 / 0.438 / 0.481	0.386 / 0.423 / 0.404	0.621 / 0.633 / 0.706
First error = point of no return	0.634 / 0.706 / 0.600	0.568 / 0.531 / 0.556	0.477 / 0.521 / 0.489	0.455 / 0.500 / 0.588
First error = first drop above clean floor	0.390 / 0.471 / 0.467	0.216 / 0.250 / 0.296	0.205 / 0.239 / 0.277	–

Restart − first error keeps its sign and significance at every threshold for every model: positive for the three Qwen models and null for Llama. The clean-floor definition is unavailable for Llama (no clean traces were run).

D Two more propositions on mitigation

Proposition 2 (the only advantage of a later restart point is shorter attempts). Let attempts from resampling point k succeed independently with probability v_k and each consume c_k generated tokens, so that a budget of B tokens buys $\lfloor B/c_k \rfloor$ attempts

and $P_k(B) = 1 - (1 - v_k)^{\lfloor B/c_k \rfloor}$. If $v_k \le v_0$, then for every B,

$$P_k(B) \;\le\; P_0\Big(B\,\tfrac{c_0}{c_k}\Big)\,.$$

Proof. $\lfloor B/c_k \rfloor = \lfloor (Bc_0/c_k)/c_0 \rfloor$ and $1-(1-v)^N$ is increasing in v, so $P_k(B) \le 1-(1-v_0)^{\lfloor (Bc_0/c_k)/c_0 \rfloor} = P_0(Bc_0/c_k)$.□

By Proposition 1, $v_\tau \le v_0$ holds on average over failed on-policy traces for any resampling point chosen without looking ahead, so such a point is worth at most restart with its budget multiplied by c_0/c_τ. Measured continuation lengths give $c_\tau/c_0 \approx 0.7$–0.8 (`results/chokebench/summary.md`), so the best such point is worth at most restart with 1.25–1.4 times the budget, before the cost of finding it.

Proposition 3 (a restart loop is exactly as faithful as its detector). For one input, let draws $Y_0, Y_1, \ldots$ be independent, each hallucinated with probability h, and let a detector flag each draw independently of the others, missing a hallucinated draw with probability β and flagging a faithful one with probability α. The loop returns the first of at most $K+1$ draws that the detector passes, and abstains otherwise. Let $q = h(1-\beta) + (1-h)\alpha$ be the flag probability. Then

$$P(\text{answer}) = 1 - q^{K+1}, \qquad P(\text{hallucinated} \mid \text{answer}) = \pi := \frac{h\beta}{h\beta + (1-h)(1-\alpha)}, \qquad \mathbb{E}[\text{draws}] = \frac{1-q^{K+1}}{1-q}.$$

Proof. Each draw is, independently, passed and hallucinated with probability $a = h\beta$, passed and faithful with probability $f = (1-h)(1-\alpha)$, or flagged with probability $q = 1-a-f$. The loop returns draw i exactly when draws $0, \ldots, i-1$ are flagged and draw i is passed, so $P(\text{hallucinated answer}) = \sum_{i=0}^{K} q^i a = a\frac{1-q^{K+1}}{1-q}$ and $P(\text{answer}) = (a+f)\frac{1-q^{K+1}}{1-q} = 1-q^{K+1}$. Their ratio is $a/(a+f) = \pi$, which does not depend on K. The loop makes draw i only if draws $0, \ldots, i-1$ were flagged, so $\mathbb{E}[\text{draws}] = \sum_{i=0}^{K} q^i$. □

Consequences. (i) A larger retry budget raises coverage, never faithfulness: the delivered hallucination rate is the detector's $P(\text{hallucinated} \mid \text{pass})$, measurable on labelled drafts. (ii) The expected cost is at most $1/(1-q)$ draws, so a detector that flags a fraction q of drafts costs at most $1/(1-q)$ generations per answer. (iii) A grader whose errors coincide with the detector's sees no hallucination in delivered answers at all, whatever π is; a grader that shares the detector's training data will typically lie between that and the truth, which is the evaluator effect of Section 9. (iv) Choosing *where* to resample does not enter π; only the detector does.